Source Code

```
#include <Adafruit_MPU6050.h>
#include <Adafruit_Sensor.h>
#include <Wire.h>
Adafruit_MPU6050 mpu;
void setup(void)
{
Serial.begin(9600);
while (! Serial)
delay(10); // will pause Zero, Leonardo, etc until serial console opens
Serial.println("Smart Glove for Sign Language");
// Try to initialize!
if (!mpu.begin())
{
Serial.println("Failed to find MPU6050 chip");
while (1)
{
delay(10);
}
}
//  Serial.println("MPU6050 Found!");
mpu.setAccelerometerRange(MPU6050_RANGE_8_G);
//  Serial.print("Accelerometer range set to: ");
switch (mpu.getAccelerometerRange())
{
case MPU6050_RANGE_2_G:
//    Serial.println("+-2G");
break;
case MPU6050_RANGE_4_G:
//    Serial.println("+-4G");
break;
case MPU6050_RANGE_8_G:
```

```
//    Serial.println("+-8G");
break;
case MPU6050_RANGE_16_G:
//    Serial.println("+-16G");
break;
}
mpu.setGyroRange(MPU6050_RANGE_500_DEG);
//  Serial.print("Gyro range set to: ");
switch (mpu.getGyroRange())
{
case MPU6050_RANGE_250_DEG:
//    Serial.println("+- 250 deg/s");
break;
case MPU6050_RANGE_500_DEG:
//    Serial.println("+- 500 deg/s");
break;
case MPU6050_RANGE_1000_DEG:
//    Serial.println("+- 1000 deg/s");
break;
CaseMPU6050_RANGE_2000_DEG:
//    Serial.println("+- 2000 deg/s");
break;
}
mpu.setFilterBandwidth(MPU6050_BAND_5_HZ);
//  Serial.print("Filter bandwidth set to: ");
switch (mpu.getFilterBandwidth())
{
case MPU6050_BAND_260_HZ:
//    Serial.println("260 Hz");
break;
case MPU6050_BAND_184_HZ:
//    Serial.println("184 Hz");
break;
case MPU6050_BAND_94_HZ:
```

```
//    Serial.println("94 Hz");
break;
case MPU6050_BAND_44_HZ:
//    Serial.println("44 Hz"); break;
case MPU6050_BAND_21_HZ:
//    Serial.println("21 Hz"); break;
case MPU6050_BAND_10_HZ:
//    Serial.println("10 Hz"); break;
case MPU6050_BAND_5_HZ:
//    Serial.println("5 Hz");
break;
}
//  Serial.println(""); delay(100);
}
void loop()
{
//  sensors_event_t a, g, temp;
//  mpu.getEvent(&a, &g, &temp);
//  /* Print out the values */
//  Serial.print("Acceleration X: ");
//  Serial.print(a.acceleration.x);
//  Serial.print(", Y: ");
//  Serial.print(a.acceleration.y);
//  Serial.print(", Z: ");
//  Serial.print(a.acceleration.z);
//  Serial.println(" m/s^2");
//  Serial.print("Rotation X: ");
//  Serial.print(g.gyro.x);
//  Serial.print(", Y: ");
//  Serial.print(g.gyro.y);
//  Serial.print(", Z: ");
//  Serial.print(g.gyro.z);
//  Serial.println(" rad/s");
//  Serial.print("Temperature: ");
```

```
//  Serial.print(temp.temperature);
//  Serial.println(" degC");
//  Serial.println("");
//  delay(1000);
//  int flux1val=analogRead(A0);
//  int flux2val=analogRead(A1);
//  int flux3val=analogRead(A2);
//  int flux4val=analogRead(A3);
//  Serial.println("Flux_1:"+String(flux1val));
//  Serial.println("Flux_2:"+String(flux2val));
//Serial.println("Flux_3:"+String(flux3val)); //Serial.println("Flux_4:"+String(flux4val));
//  delay(1000);
sensors_event_t a, g, temp; mpu.getEvent(&a, &g, &temp);
float ax = a.acceleration.x;
float ay = a.acceleration.y;
float az = a.acceleration.z;
//  Serial.print("Acceleration X: ");
//  Serial.print(ax);
//  Serial.print("  Y: ");
//  Serial.print(ay);
//  Serial.print("  Z: ");
//  Serial.println(az);
float threshold = 3.0; // Sensitivity for tilt detection
if (ax > threshold && abs(ay) < threshold)
{
Serial.println('K');
delay (1200);
}
else if (ax < -threshold && abs(ay) < threshold)
{
Serial.println('L');
delay (1200);
}
else if (ay > threshold && abs(ax) < threshold)
```

```
{
Serial.println('M');
delay(1200);
}
else if (ay < -threshold && abs(ax) < threshold)
{
Serial.println('N');
delay(1200);
}
Else
{
//    Serial.println("Stable");
// Optional: when no clear tilt
}
//  Serial.println();
// ---- Flux sensors (unchanged) ----
int flux1val = analogRead(A0);
int flux2val = analogRead(A1);
int flux3val = analogRead(A2);
int flux4val = analogRead(A3);
//  Serial.println("Flux_1:" + String(flux1val));
//  Serial.println("Flux_2:" + String(flux2val));
//  Serial.println("Flux_3:" + String(flux3val));
//  Serial.println("Flux_4:" + String(flux4val));
//  Serial.println("----------------------");
delay(1000);
if(flux1val<400)
{
//    Serial.println("Can u repeat again");
Serial.println('A');
delay(1200);
}
if(flux2val<400)
{
```

```
//    Serial.println("Need to Go for Hospital");
Serial.println('B');
delay(1200);
}
if(flux3val<400)
{
//    Serial.println("Hello!! Have a nice day");
Serial.println('C');
delay(1200);
}
if(flux4val<400)
{
//    Serial.println("Bye Takecare");
Serial.println('D');
delay(1200);
}
if(flux1val<400 && flux2val<400)
{
Serial.println('E'); delay(1200);
}
if(flux1val<400 && flux3val<400)
{
Serial.println('F'); delay(1200);
}
if(flux1val<400 && flux4val<400)
{
Serial.println('G'); delay(1200);
}
if(flux2val<400 && flux3val<400)
{
Serial.println('H');
delay(1200);
}
if(flux2val<400 && flux4val<400)
```

```
{
Serial.println('I');
delay(1200);
}
if(flux3val<400 &&flux4val<400)
{
Serial.println('J');
delay(1200);
}
```